

Elevation, oxygen, and the origins of viviparity

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Abstract

Research focused on understanding the evolutionary factors that shape parity mode evolution among vertebrates have long focused on squamate reptiles (snakes and lizards), which contain all but one of the evolutionary transitions from oviparity to viviparity among extant amniotes. While most hypotheses have focused on the role of cool temperatures in favoring viviparity in thermoregulating snakes and lizards, there is a growing appreciation in the biogeographic literature for the importance of lower oxygen concentrations at high elevations for the evolution of parity mode. However, the physiological mechanisms underlying how hypoxia might reduce fitness, and how viviparity can alleviate this fitness decrement, has not been systematically evaluated. We qualitatively evaluated previous research on reproductive and developmental physiology, and found that (1) hypoxia can negatively affect fitness of squamate embryos, (2) oxygen availability in the circulatory system of adult lizards can be similar or greater than an egg, and (3) gravid females can possess adaptive phenotypic plasticity in response to hypoxia. These findings suggest that the impact of hypoxia on the development and physiology of oviparous and viviparous squamates would be a fruitful area of research for understanding the evolution of viviparity. To that end, we propose an integrative research program for studying hypoxia and the evolution of viviparity in squamates.

KEYWORDS

elevation, hypoxia, viviparity

1 | BACKGROUND

There are two basic reproductive modes in animals, oviparity and viviparity. Oviparity generally describes when the offspring develop external to the mother in an enclosed egg and is ancestral in most major animal groups, including all extant vertebrates (Griffith et al., 2015; King & Lee, 2015; Wright et al., 2015). The second reproductive strategy is viviparity, where offspring complete their development inside of the mother and emerge as free-living neonates or larvae (Amoroso, 1968; Wake, 1992). This reproductive mode evolved from oviparity many times across the vertebrate tree of life (Blackburn, 1992; Griffith et al., 2015; King & Lee, 2015; but see Pyron & Burbrink, 2015), but is perhaps most studied in the vertebrate clade of amniotes which includes mammals, birds, and non-

avian reptiles. Indeed, amniotes are named for extraembryonic tissues that are crucial for both oviparous and viviparous reproduction (Favaron et al., 2015; Ferner & Mess, 2011). While mammals are the most familiar viviparous amniotes, viviparity in mammals has only evolved a single time in the transition from the ancestors of modern monotremes (platypis and echidnas) to metatherian (marsupial) and eutherian (placental) mammals. All of the remaining 100+ extant origins of viviparity among extant amniotes occurred in the squamate reptiles, which include modern snakes and lizards (Blackburn, 1992, 2006). Previous research in squamates has noted the high frequency of viviparous species at high elevations and latitudes and concluded that viviparity is an adaptation to cold temperatures (e.g., the cold-climate hypothesis; Shine, 1983b, 1985; Tinkle & Gibbons, 1977). This hypothesis posits that (1) developmental temperature can

influence offspring viability and fitness and (2) thermoregulation by gravid females allows more precise temperature regulation for the developing embryo in utero than would be available in ovo (Shine, 1983b, 1985, 1995). However, recent research at a variety of evolutionary and spatial scales have found that cool temperatures alone are not sufficient to explain the origins of viviparity (e.g., Lambert & Wiens, 2013; Pincheira-Donoso et al., 2017; Watson et al., 2014). Rather, the evolutionary origins of viviparity tend to be concentrated at high elevations, which not only tends to be colder than low elevations, but will also be hypoxic due to lower partial pressures of all atmospheric gases (Peacock, 1998). This perspective integrates phylogenetic, geographic, and physiological evidence to discuss mechanisms whereby viviparity might be a reproductive solution to the problem of low oxygen for developing embryos.

2 | BIOGEOGRAPHY AND THE MACROEVOLUTION OF VIVIPARITY

Biologists have long recognized the relative preponderance of viviparous species of squamates at high elevations and latitudes (Gadow, 1910; Greer, 1966; Mell, 1929; Packard et al., 1977; Packard, 1966; Sergeev, 1940; Weekes, 1935). Indeed, by compiling point of observation records from iNaturalist (iNaturalist.org) accessed on August 07, 2020, we found that approximately 42% of snakes and lizards at high latitudes (above the 49th parallel) are viviparous (Figure 1), and approximately 72% at high elevations (above 3500 m) are viviparous (Figure 1), compared to the 15%–20% frequency of viviparity across all reptiles (Blackburn, 2000a; Tinkle & Gibbons, 1977). The high frequency of viviparity in snakes and lizards at high elevations and latitudes has been consistently interpreted as evidence for viviparity as a cold adaptation (Guillette et al., 1980; Sergeev, 1940; Shine & Bull, 1979; Tinkle & Gibbons, 1977; Weekes, 1933). Ultimately, these ideas became organized as the cold-climate hypothesis (or cold-adaptation hypothesis), which predicted that viviparity arose as an adaptation to the negative impacts of cool temperatures during development (Shine & Bull, 1979; Shine, 1983b, 1985; Tinkle & Gibbons, 1977). Because gravid female lizards and snakes can thermoregulate, this hypothesis suggests that gravid females can increase and regulate temperature better than is possible for an egg laid in a nest exposed to ambient temperature. Recent work studying phylogenetic relationships and geographic distributions of squamates generally supports this hypothesis (e.g., Feldman et al., 2015; Lambert & Wiens, 2013; Ma et al., 2018; Watson et al., 2014). However, some findings in research that incorporates phylogenetic relationships among organisms and detailed quantitative characterization of climate complicates this picture. Watson et al. (2014) focused analyses on recent origins of viviparity by studying reproductively bimodal lizard genera and generally found support for the cold climate hypothesis, but also found that elevation explained the distribution of viviparity better than temperature or latitude in many groups. This finding was congruent with earlier research that suggested high elevation is a stronger selective pressure than high

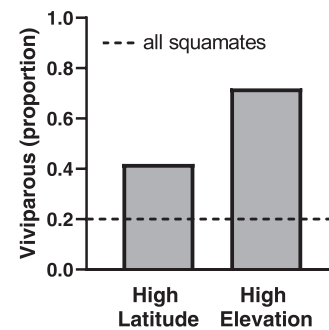


FIGURE 1 Proportion of viviparous squamates at high latitude (above 49° and below -49° latitude) and high elevation (>3500 m). The dashed line represents the proportion among all squamates (~20%) according to Blackburn (2000a)

latitude among the lizard genus *Sceloporus* (Guillette et al., 1980). Hodges (2004) and Lambert and Wiens (2013) also found evidence that the evolution of viviparity in the lizard family Phrynosomatidae is most common at high elevations but at lower latitudes, partially contradicting the cold-climate hypothesis. Hence, the evolutionary origins of viviparity seems concentrated in high elevations at lower latitudes, which would be warmer and hence relatively oxygen-depauperate. In fact, Pincheira-Donoso et al. (2017) found that low oxygen availability and cold temperature were important factors underlying the evolution of viviparity in the lizard genus *Liolaemus*. Together, these data suggest that (1) temperature alone may not be adequate to explain the evolution of viviparity in squamate reptiles and (2) oxygen limitation might also influence the evolution of viviparity. However, while the biogeographic literature has acknowledged a potential role of oxygen limitation for influencing the evolution of viviparity, whether or not substantial literature on the role of hypoxia on physiology, development, and fitness is consistent with the hypothesis that hypoxia at high elevations exerted selection for viviparity has not been evaluated.

3 | FUNCTIONAL PREDICTIONS OF HYPOXIA AND VIVIPARITY

If environmental hypoxia at high elevations is important to the evolution of viviparity, then there should be evidence supporting several functional predictions regarding the impact of hypoxia on embryonic and maternal physiology. First, hypoxia experienced by developing embryos should negatively impact offspring viability and fitness (section 3.1). Second, oxygen availability in the circulatory system of an adult lizards at high elevations should be similar or greater than that available to an egg surrounded by ambient oxygen at high elevations (section 3.2). Finally, gravid females should exhibit physiological mechanisms that mitigate the effects of reduced atmospheric oxygen on developing embryos, such as canalization that maintains homeostasis regardless of environmental hypoxia or phenotypic plasticity that shifts circulatory and respiratory physiology to counteract low ambient oxygen at high elevations (section 3.3).

3.1 | Hypoxia should negatively impact offspring fitness

Generally, hypoxia negatively influences fitness in amniotes that complete their development in eggs. For birds, which can live at high elevations and are the largest group that is wholly oviparous, oxygen deprivation during development can negatively impact physiological function, morphological trajectories, body size, and delay development in some species (Azzam et al., 2007; Chan & Burggren, 2005; Dzialowski et al., 2002; Smith et al., 1969), although other species experience no deleterious effects associated with incubation in high elevation hypoxic environments (Carey et al., 1982; Snyder et al., 1982). Indeed, some high-elevation birds have developmental, metabolic, and structural adaptations of the developing embryo and egg to hypoxia (León-Velarde & Monge-C, 2004; León-Velarde et al., 1997). It is worth noting that the majority of the relevant research has used low-elevation birds as models.

Turtles are another solely oviparous clade (Blackburn, 2000a; Rafferty et al., 2013; Shine, 1983a) whose members may experience some level of hypoxia, with some species ranges extending into moderately high elevations (Cooley et al., 2003; Macip-Ríos et al., 2011) and other species having nests periodically inundated with water (Fordham et al., 2006; Kennett et al., 1993). This hypoxia can depress growth, metabolism, and size at hatching (Kam, 1993; Nechaeva, 2011), and turtles can arrest embryonic development in response to hypoxia (Rafferty et al., 2013). Crocodilians are also solely oviparous (Blackburn, 2000a), but only exist at lower elevations. However, some species like American alligators (*Alligator mississippiensis*) lay eggs in nests that can become hypoxic when periodically inundated with water and exhibit a high degree of physiological and morphological plasticity under different oxygen levels (Eme et al., 2011). Under hypoxic conditions, alligators have been shown to exhibit increased relative embryonic cardiac size (Crossley & Altimiras, 2005; Crossley et al., 2017) and increased embryonic hematocrit (Warburton et al., 1995), but body size at hatching and metabolism are still negatively impacted by chronic hypoxia (Owerkowicz et al., 2009). The relative cardiac size reported in Crossley and Altimiras (2005) was a result of this decreased body size rather than an increase in cardiac size. This functionally provides a relatively larger cardiovascular system for a smaller organism. Thus, for taxonomic groups that are only oviparous, hypoxia can reduce fitness and there are a variety of adaptations that may mitigate the fitness consequences of low oxygen availability that we do not discuss here.

Squamates are one of only two major clades of extant amniotes that are reproductively bimodal, and existing literature suggests that they exhibit significant compensatory reactions during development to mitigate negative impacts of hypoxia. We surveyed the existing literature for the impact of hypoxia on development in squamate reptiles by searching “snake embryo hypoxia,” “snake embryo oxygen,” “lizard embryo hypoxia,” “lizard embryo oxygen,” “squamate embryo oxygen,” and “squamate embryo hypoxia” and examining the first 100 results on both Google Scholar and ISI Web of Science on

TABLE 1 Impacts of hypoxia on developing squamate embryos in ovo

Impact of hypoxia	Species	Citation
Compensatory		
Cardiac hypertrophy	<i>Podarcis muralis</i>	(Cordero et al. 2017)
Increased heart rate	<i>Acritoscincus duperreyi</i>	(Du et al. 2010)
	<i>Natrix maura</i>	(Souchet et al. 2020)
	<i>Podarcis muralis</i>	(Kouyoumdjian et al. 2019)
Deleterious		
Developmental malformations	<i>Anolis sp.</i>	(Nunez & Sanger 2019)
Reduced cognition	<i>Eremias argus</i>	(Sun et al. 2014)
Reduced hatchling mass	<i>Natrix maura</i>	(Souchet et al. 2020)
	<i>Podarcis muralis</i>	(Cordero et al. 2017)
	<i>Sceloporus olivaceous</i>	(Pérez Guerra 2019)
	<i>Sceloporus virgatus</i>	(Parker & Andrews 2006)
	<i>Scincella lateralis</i>	(Flewelling & Parker 2015)
Reduced heat tolerance	<i>Urosaurus ornatus</i>	(Parker & Andrews 2006)
	<i>Sceloporus tristichus</i>	(Smith et al. 2015)
Reduced performance	<i>Antaresia childreni</i>	(Stahlschmidt & Denardo 2009)
	<i>Natrix maura</i>	(Souchet et al. 2020)

September 27th, 2020 (first four search terms) and May 9th, 2021 (last two search terms). Rather than attempting to be exhaustive, our goal was to capture a general understanding of how hypoxia impacts development in squamates (Table 1). We found very few new papers that were relevant past the first forty results of each search and methods, and the specific papers returned by each method overlapped extensively.

Hypoxia during development of reptilian embryos in ovo can induce compensatory physiological and morphological reactions such as an elevated heart rate and cardiac hypertrophy (e.g., *Acritoscincus duperreyi*: Du et al., 2010; *Podarcis muralis*: Kouyoumdjian et al., 2019; *Natrix maura*: Souchet et al., 2020). Other deleterious effects of hypoxia on embryos can include reduced heat tolerance (Smith et al., 2015), reduced performance (Souchet et al., 2020), and increased likelihood of developmental malformations (Nunez & Sanger, 2019). Oviparous wall lizards (*Podarcis muralis*) suppress metabolism and experience cardiac hypertrophy to maintain developmental rate at high elevations, but hatchlings incubated under hypoxic condition still

have a lower relative mass (Cordero et al., 2017). Reduced mass for embryos in hypoxia may occur because development occurs at a normal rate but with diminished tissue anabolism (Flewelling & Parker, 2015). Under hypoxic conditions induced by nest guarding in pythons, hatchlings are smaller and exhibit reductions in fitness related traits, such as strength and speed (Stahlschmidt & Denardo, 2009). The reduced offspring size caused by hypoxia during development has important ramifications for fitness, as offspring size is generally tightly associated with fitness across taxa (Einum & Fleming, 2000; Janzen, 1993; Smith & Fretwell, 1974). Among squamates, body size of neonates is positively associated with fitness (Ferguson & Fox, 1984; Uller & Olsson, 2010; Webb et al., 2006), although this association may be conditional and vary among populations, with time of year, and sex (Langkilde & Shine, 2005; Sinervo et al., 1992; Warner & Shine, 2007).

Beyond offspring size, embryos of a low-elevation population of the Mongolian racerunner lizard (*Eremias argus*) experience no apparent deleterious effects of hypoxia when transplanted to higher elevations (Li et al., 2020) yet exhibit a longer incubation time and reduced cognition (Sun et al., 2014). Intriguingly, there were no differences in survival, differentiation, or growth between early-development eastern fence lizards (*Sceloporus undulatus*) embryos retained inside the mother under hypoxic and normoxic conditions (Parker et al., 2004), which is consistent with the idea that the effects of hypoxia may be mitigated by the maternal environment even in oviparous species. These studies indicate that hypoxic conditions during development can negatively impact fitness-related traits and thereby exert strong selection for a physiological solution for the problem of low oxygen at high elevations.

3.2 | Oxygen availability could be higher in utero than in ovo

While hypoxia might have a negative impact on developing embryos, does the environment in utero offer a more normoxic environment than the ambient environment for the developing embryos? Andrews (2002) makes the argument that (1) hypoxia is detrimental to developing embryos and (2) a fluid medium, such as that in the maternal environment, would be relatively oxygen-depauperate compared to ambient air (Andrews, 2002; Jackson, 2007). Hence, viviparity would not evolve unless embryo retention was accompanied by physiological and morphological adaptations that enhance oxygen availability above that of ambient air (Andrews, 2002; Jackson, 2007). Additionally, cooler temperatures may mitigate the acute effects of hypoxia by reducing metabolic rate of ectothermic organisms. These two factors together may explain the relative abundance of viviparous species at high elevation (Figure 1).

However, the internal environment of the egg is also aqueous and should therefore experience a reduction in the partial pressure of respiratory gases compared to the ambient environment similar in effect to the environment in utero, disregarding maternal circulation and respiration. Hence, the evolution of viviparity should be

preceded by adaptations, such as reduction of eggshells and vascularization of the extraembryonic and oviductal membranes (Andrews & Mathies, 2000; Andrews, 2002). Extending Fick's Principle (Fick, 1870) to this system, the rate of oxygen uptake by an organ or tissue, in this case an embryo, is directly related to bloodflow, oxygen concentration in the environment, and the efficiency of the organ in obtaining oxygen from the environment. Each of these variables can be enhanced in the form of morphological and physiological adaptations through selection.

As with virtually all vertebrates, the presence of oxygen-binding hemoglobin, negative-pressure ventilation, and convective gas exchange in the lungs of fully formed adults would raise the potential oxygen saturation (and, therefore, the partial pressure of oxygen) within the internal environment relative to eggs. While the presence of hemoglobin and convective gas exchange also occur in ovo (Sartori et al., 2018), it would not likely approach the effectiveness of the fully formed and actively ventilated adult lungs and associated circulatory system, especially early in development. We canvassed the existing literature for published blood oxygen capacity in squamate reptiles by searching "lizard blood oxygen" or "snake blood oxygen" and scrutinizing the first 100 results on both Google Scholar and ISI Web of Science on October 9th, 2020. As with the previously described search, rather than attempting to be exhaustive, our goal was to capture a representative sample of published blood oxygen levels of squamates (Table 2). We found that oxygen concentration of squamate blood ranges from 3.6 to 16.2 ml·100 ml⁻¹ (% volume), while fluid similar to blood plasma lacking hemoglobin (estimated as water at sea level or 1 atmosphere, 20°C, and 0.9% salinity which is similar to blood plasma), would theoretically have a maximum dissolved oxygen value of approximately 1.2 ml·100 ml⁻¹ (Figure 2). Even across a broad range of PCO₂, temperature, and pH, PO₂ would still be dramatically higher in blood than most regions of the egg because of the respiratory pigment hemoglobin (Randall et al., 2002; Roughton, 1964). While we recognize that the egg is not avascular and that there are mechanisms that can increase oxygen availability to embryos in an egg, our example illustrates that the partial pressure of oxygen in the bloodstream of reptiles is variable and higher than what is available in a similar aqueous environment. Without even invoking plasticity or adaptation, the maternal environment could feasibly provide developing embryos with an enhanced oxygen environment greater than would be available for eggs even at low elevations, as illustrated by the fact that viviparous species exist in normoxic conditions.

Beyond maternal physiology, extraembryonic tissues facilitate maternal-embryonic gas exchange in viviparous species. The chorioallantoic placenta in viviparous lizards exists along a morphological spectrum from simple to complex, with some viviparous lizards even possessing a chorioallantoic placenta similar to the true placenta of eutherian mammals which could increase maternal-embryonic gas and nutrient exchange (Blackburn & Vitt, 2002; Blackburn, 1992, 2000b, 2015; Stewart, 2015). Viviparous spiny lizards (*Sceloporus bicanthalis*) have a better-developed chorioallantoic placenta and more vascularized uterine walls than oviparous sister species

TABLE 2 Published oxygen capacity of blood at temperatures and elevation for squamates

Species	Oxygen capacity (volume %)	Temperature (°C)	Elevation (m)	Citation
Lizards-Oviparous				
<i>Acanthodactylus erythrurus</i>	4.7			(Pough 1979)
<i>Aspidoscelis inornatus</i>	10.8	39.3	1646–1981	(Dawson & Poulson 1962)
<i>Aspidoscelis sackii</i>	11.6	39.4	1310–1402	(Dawson & Poulson 1962)
<i>Aspidoscelis tigris</i>	9.6	41.3	1402	(Dawson & Poulson 1962)
<i>Cophosaurus texanus</i>	7.8	38.2	1402	(Dawson & Poulson 1962)
<i>Crotaphytus collaris</i>	10.4	37.8	610	(Dawson & Poulson 1962)
<i>Dipsosaurus dorsalis</i>	9.6	42.1	183	(Dawson & Poulson 1962)
<i>Dipsosaurus dorsalis</i>	9.3	30–35		(Pough 1976)
<i>Elgaria multicarinata</i>	12.6	25.8	183	(Dawson & Poulson 1962)
<i>Elgaria multicarinata</i>	6.7	15–30		(Pough 1976)
<i>Gambelia wislizeni</i>	10.4	38.9	610	(Dawson & Poulson 1962)
<i>Iguana iguana</i>	7.9	20–35		(Pough 1976)
<i>Iguana iguana</i>	8.4	28–32		(Tucker 1966)
<i>Iguana iguana</i>	10.5			(Wood & Moberly 1970) ^a
<i>Paralaudakia caucasia</i>	11.7			(Verjbiinskaya 1944) ^a
<i>Phrynosoma cornutum</i>	9.3	37.3	1646	(Dawson & Poulson 1962)
<i>Phrynosoma modestum</i>	12.1		1402	(Dawson & Poulson 1962)
<i>Psammodromus algirus</i>	4			(Pough, 1979)
<i>Pseudopus apodus</i>	9.8			(Verjbiinskaya 1944) ^a
<i>Sauromalus hispidus</i>	9.7	25		(Bennett 1973) ^a
<i>Sauromalus obesus</i>	9.8	30–35		(Pough 1976)
<i>Sauromalus obesus</i>	10.8			(Pough 1979)
<i>Sceloporus clarki</i>	8.3		1554	(Dawson and Poulson 1962)
<i>Sceloporus graciosus</i>	10.9		183	(Dawson & Poulson 1962)
<i>Sceloporus magister</i>	8.9	30		(Pough 1976)
<i>Sceloporus magister</i>	10.9			(Ryerson 1949) ^a
<i>Sceloporus occidentalis</i>	9.5	33.8	183	(Dawson & Poulson 1962)
<i>Sceloporus occidentalis</i>	5.8	20–35		(Pough 1976)
<i>Sceloporus orcutti</i>	8	25–30		(Pough 1976)
<i>Sceloporus virgatus</i>	9.6	34.8	1554–1707	(Dawson & Poulson 1962)
<i>Tupinambis merianae</i> Summer)	13.1	25		(Andrade et al. 2004)
<i>Tupinambis merianae</i> (Winter)	11.02	17		(Andrade et al. 2004)
<i>Tupinambis teguixin</i>	9.5	35		(Pough 1976)
<i>Uta stansburiana</i>	8.7	35–38	1402	(Dawson & Poulson 1962)
<i>Uta stansburiana</i>	9.2	20–35		(Pough 1976)
<i>Varanus exanthematicus</i>	10.1			(Wood et al. 1977) ^a
<i>Varanus gouldi</i>	8	25		(Bennett 1973) ^a
<i>Varanus niloticus</i>	9.3	25		(Wood & Johansen 1974)

(Continues)

TABLE 2 (Continued)

Species	Oxygen capacity (volume %)	Temperature (°C)	Elevation (m)	Citation
Lizards-Viviparous				
<i>Elgaria coerulea</i>	6.8			(Pough 1979)
<i>Phrynosoma douglassi</i>	10.2	36	1646	(Dawson & Poulson 1962)
<i>Sceloporus jarrovi</i>	7.7	34.8	1554	(Dawson & Poulson 1962)
<i>Sceloporus jarrovi</i>	8.3	34.8	2621	(Dawson & Poulson 1962)
<i>Sceloporus poinsetti</i>	10	34.2	1341	(Dawson & Poulson 1962)
Snakes-Oviparous				
<i>Drymarchon corais</i>	4.2			(Pough 1979)
<i>Lampropeltis getula</i>	4.5			(Pough 1979)
<i>Lampropeltis triangulum</i>	8.7	20–25		(Pough 1976)
<i>Laticauda colubrina</i>	11			(Pough 1979)
<i>Laticauda colubrina</i>	10.7	28		(Pough & Lillywhite 1984)
<i>Laticauda columbrina</i>	11.04	30		(Feder 1980)
<i>Laticauda columbrina</i>	9.3	27		(Seymore 1976)
<i>Pantherophis guttatus</i>	4.2			(Pough 1979)
<i>Pantherophis obsoletus</i>	6.7			(Pough 1979)
<i>Pituophis melanoleucus</i>	9.3			(Greenwald 1971) ^a
<i>Pituophis melanoleucus</i>	11.4	20–25		(Pough 1976)
<i>Python molurus</i>	5.7			(Pough 1979)
<i>Python reticulatus</i>	7.1			(Pough 1979)
<i>Xenopeltis unicolor</i>	10			(Pough 1979)
Snakes-Viviparous				
<i>Acalyptophis peronii</i>	7.9	28		(Pough & Lillywhite 1984)
<i>Acrocordus granulatus</i>	16.2	30		(Feder 1980)
<i>Acrocordus javanicus</i>	9.3			(Johansen & Lenfant 1972) ^a
<i>Agkistrodon piscivorous</i> (Fetal)	7.1	23		(Birchard et al. 1984a)
<i>Agkistrodon piscivorous</i> (Maternal)	7.6	23		(Birchard et al. 1984a)
<i>Aipysurus laevis</i>	12.7	28		(Pough & Lillywhite 1984)
<i>Boa constrictor</i>	13.5			(Johansen & Lenfant 1972) ^a
<i>Cerberus rhynchops</i>	3.6	30		(Feder 1980)
<i>Emydocephalus annulatus</i>	6.53	30		(Feder 1980)
<i>Emydocephalus annulatus</i>	6.5			(Pough 1979)
<i>Eryx johnei</i>	6.4			(Pough 1979)
<i>Hydrophis melanocephalus</i>	10.8	28		(Pough & Lillywhite 1984)
<i>Nerodia rhombifer</i>	7.6			(Pough 1979)
<i>Nerodia sipedon</i>	11.1	20–25		(Pough 1976)
<i>Pelamis platura</i>	10.2	28		(Pough & Lillywhite 1984)
<i>Pseudechis porphyriacus</i>	10.9	27		(Seymour 1976)
<i>Thamnophis couchi</i>	9			(Pough 1979)

TABLE 2 (Continued)

Species	Oxygen capacity (volume %)	Temperature (°C)	Elevation (m)	Citation
<i>Thamnophis sirtalis</i> (Males)	9.6			(Birchard et al. 1984b)
<i>Thamnophis sirtalis</i> (Nongravid females)	9.2			(Birchard et al. 1984b)
<i>Thamnophis sirtalis</i> (Gravid Females)	11.1			(Birchard et al. 1984b)
<i>Thamnophis sirtalis</i> (Post-partum females)	9.7			(Birchard et al. 1984b)
<i>Thamnophis sirtalis</i>	9.5	10–30		(Pough 1976)
<i>Vipera berus</i>	12.3	5		(Johansen & Lykkeboe 1979)
<i>Vipera berus</i>	9.9	25		(Johansen & Lykkeboe 1979)

^aSourced directly from Pough (1979).

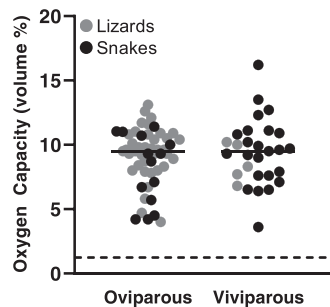


FIGURE 2 Published values of blood oxygen capacity for oviparous and viviparous squamates (see Table 2). The dashed line represents the approximate maximum dissolved oxygen of 20°C water at sea level and salinity similar to blood plasma (0.9%) which we use here as a null comparison

(Guillette & Jones, 1985). Similarly, uterine and placental angiogenesis has been recorded in viviparous Australian skinks (Murphy et al., 2010b; Parker et al., 2010). Such angiogenesis can be upregulated in response to increased oxygen demand by the embryo (Parker et al., 2006) and is potentially regulated by hormones (Murphy et al., 2010a; Whittington et al., 2015). In these species, significant amounts of nutrients and respiratory gases are shared between the mother and the developing embryo (Thompson & Speake, 2006). The exchange of gases could be made more efficient by increasing the oxygen affinity of embryonic tissues relative to maternal tissues. This difference in oxygen affinity between maternal and fetal blood is common among reptiles (Blackburn, 2000b; Pough, 1980) and is present in most vertebrates where these values have been measured (Ingermann, 1992). Gravid viviparous snakes can reduce maternal hemoglobin oxygen affinity below that of embryos so that embryos have a relatively higher oxygen affinity (Holland et al., 1990; Ingermann, 1992). Similarly, the oxygen affinity of embryonic blood is generally higher than maternal blood in many species of squamates (e.g. Lizards: Blackburn, 1993, 2000b; Grigg & Harlow, 1981; Snakes: Ingermann, 1992; Ragsdale & Ingermann, 1991). These differences in oxygen affinity can result in the embryonic and maternal blood both achieving saturation if the blood oxygen capacity of both mother and embryo are the same as reported in *Agkistrodon piscivorus* by

Birchard et al. (1984a). Taken together, these studies suggest that by coupling this increased oxygen capacity of hemoglobinated blood with higher affinity for oxygen in fetal blood compared to maternal blood (Ingermann, 1992), the maternal environment could have similar or higher oxygen partial pressures than an egg would experience in that same ambient environment. However, viviparous lizards can also exhibit adaptive physiological plasticity in hypoxic environments to further increase oxygen availability in utero.

3.3 | Physiological plasticity in response to hypoxia

While many organisms have adapted resilient phenotypes that canalize homeostasis and allow them to persist in otherwise inhospitable circumstances, others rely on physiological plasticity to mitigate the effects of relatively harsh environments (Debat & David, 2001; Van Buskirk & Steiner, 2009). The costs of canalization and phenotypic plasticity are typically mild and may only result in significant reduction in fitness under extreme circumstances (Van Buskirk & Steiner, 2009). Therefore, we expect the trade-off of maintaining physiological plasticity to benefit the organism unless faced with conditions that would otherwise favor strong selection. In those situations, strong selection may favor additional adaptations to mitigate the effects of harsh environments.

Both reversible and irreversible (i.e., canalization) physiological plasticity of the mother in response to high-elevation hypoxic environments may also allow a viviparous squamate to meet the oxygen demands of a developing embryo, compared to an embryo in ovo. Theoretically, the gravid female can increase oxygen available to the developing embryo in utero by increasing maternal hematocrit (increasing oxygen carrying capacity of maternal blood), maternal respiration (increasing oxygen content of maternal blood), maternal heart rate (increasing blood flow to maternal lungs and oviduct), oviduct and placental vascularization (increasing surface area for oxygenated blood contact with placental membranes), and relative fetal blood oxygen affinity (facilitating efficient oxygen acquisition from the maternal blood) (Figure 3). These phenomena are most thoroughly studied in mammals. In response to acute exposure to hypoxia, viviparous mammals exhibit the hypoxic ventilatory

response (Moore, 2000; Teppema & Dahan, 2010) which meets oxygen demands by increasing tidal volume and respiratory rate. Chronic exposure to hypoxic environments can induce an increase in hemoglobin concentration and oxygen capacity of the blood in viviparous mammals (Akunov et al., 2018). Among gravid mammals, these adaptations are accompanied by increased vascularization of the maternal and fetal membranes, resulting in increased blood flow and oxygen delivery to the embryo (Krebsa et al., 1997; Postigo et al., 2009). If similar adaptations are also present in squamates, then viviparity might confer a fitness benefit compared to oviparity in a hypoxic environment at high elevations (Figure 3).

Indeed, previous research has documented that squamate reptiles can also alter physiology in response to hypoxia. Gravid female spiny lizards (*Sceloporus grammicus* and *S. torquatus*) increase ventilation at high elevations when compared to males and non-gravid

females of the same species (Pérez Guerra, 2019). The viviparous shingleback skink (*Tiliqua rugosa*) does not alter respiration during pregnancy in normoxic conditions (Munns & Daniels, 2007), indicating that this behavior may only be required when environmental PO_2 is reduced. Beyond ventilation, mass-specific oxygen consumption is higher in gravid female lizards than expected based on allometric scaling than non-gravid females (Beuchat & Vleck, 1990; Guillelte, 1982), presumably reflecting energy demands of the tissue anabolism in the developing embryo. Kouyoumdjian et al. (2019) reported the capacity of gravid wall lizards (*Podarcis muralis*) to increase hemoglobin concentrations when transplanted to higher elevation sites, thereby increasing the oxygen carrying capacity of the maternal blood. Together, these studies suggest that adaptive physiological plasticity of gravid squamates in response to hypoxic environments at high elevations and increased oxygen demands of the developing

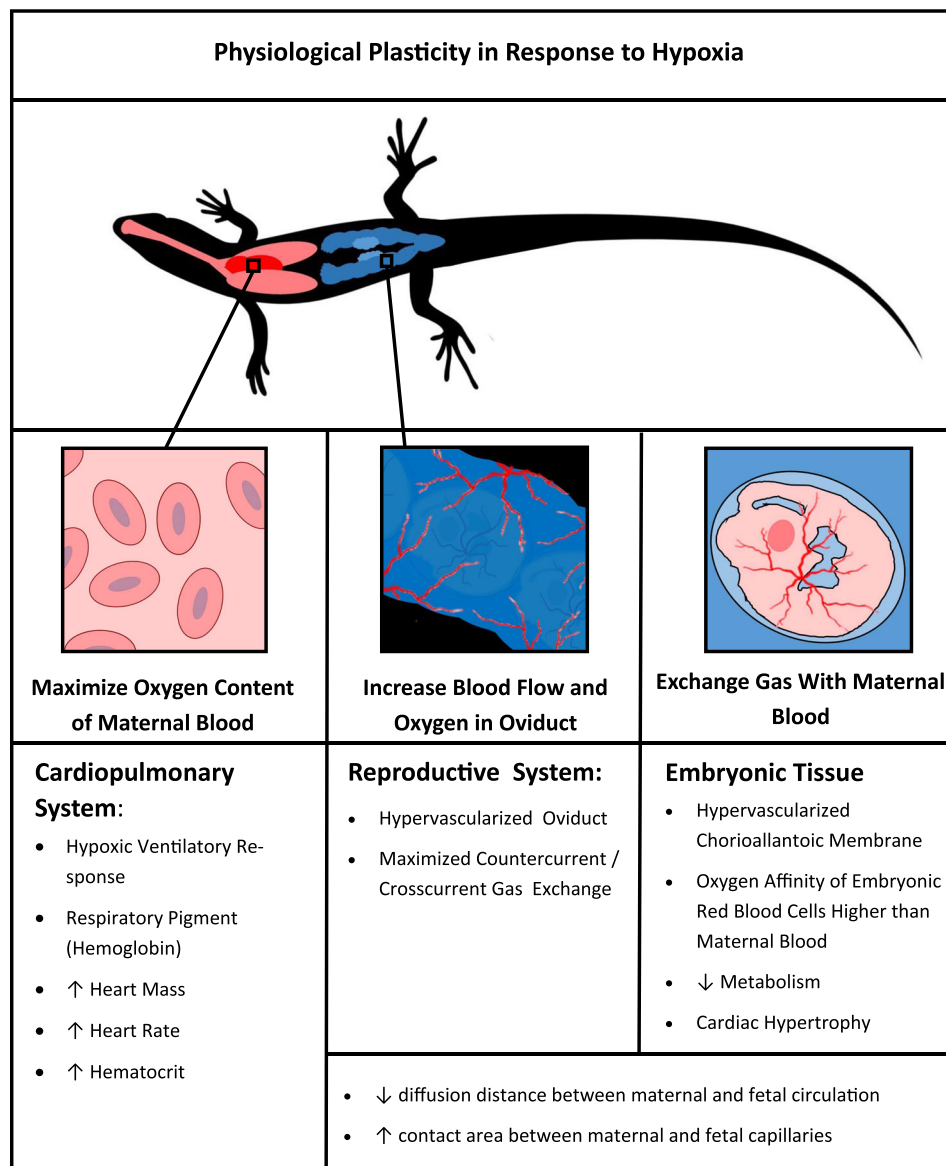


FIGURE 3 Requirements for viviparity and the physiological and morphological adaptations that facilitate these requirements and mitigate the effects of hypoxia on developing embryos

embryo could provide a more favorable oxygen environment in utero than in ovo.

4 | AN INTEGRATIVE PROGRAM FOR STUDYING HYPOXIA AND THE EVOLUTION OF VIVIPARITY

While there is substantial evidence that hypoxia plays a role in the evolution of viviparity, there is little research testing how oxygen availability actually impacts physiology and fitness of embryos and adults of both oviparous and viviparous reptiles. We have identified five areas as promising directions for future research to describe the developmental, respiratory, and cardiovascular physiology of viviparous and oviparous species during hypoxia. If hypoxia is indeed important in the evolution of viviparity, then we predict mechanisms that maximize gas exchange to be enhanced and the effects of hypoxia on developing embryos will be reduced in viviparous species compared to oviparous species.

4.1 | Studying viviparous species

While the impact of hypoxia on development in squamates has been relatively well studied, this study almost exclusively focused on oviparous species. Future research should also test the impact of hypoxia on embryonic development in viviparous species. The results from this study would reveal whether viviparity ameliorates the fitness decrement of hypoxia on developing squamates.

4.2 | Measuring the developmental environment

Little is known about actual oxygen concentration of the developmental environment of oviparous and viviparous squamate embryos. Future research should directly measure oxygen concentration of embryos in ovo and in utero, as well as oxygen concentration of maternal circulatory and reproductive systems of viviparous species and nests of oviparous species under hypoxia and normoxia. These studies could ascertain whether oxygen availability in the developmental environment actually differs between viviparous and oviparous species, which has implications for how hypoxia could favor the evolution of viviparity.

4.3 | Phenotypic plasticity

Whether the reproductive tracts of viviparous squamate species can exhibit adaptive phenotypic plasticity in response to hypoxia is also relatively unknown. Future research on the histology and anatomy of the reproductive tract of viviparous species should evaluate how hypoxia affects morphology and physiology related to gas exchange, including morphology of the embryonic membranes. Such studies

should not be limited to descriptive morphology but should also include quantification of the potential gas exchange surface area, diffusion distance, and blood capillary volume among other relevant measures. This study would be crucial for determining how reproductive tracts of viviparous species can respond to low oxygen to provision developing offspring with adequate oxygen.

4.4 | Cardio-respiratory regulation

Another potentially plastic response that warrants additional research relates to how gravid females of viviparous species can alter respiration and circulation in hypoxic conditions. Studies of viviparous species should compare respiratory (e.g., respiration rate, inspiration volume) and circulatory (hemoglobin concentration, vascularization) physiology under hypoxia and normoxia. This would reveal whether adaptive plasticity of maternal physiology and behavior could provision developing offspring with adequate oxygen given environmental hypoxia.

4.5 | Embryonic response to hypoxia

Finally, the genetic and physiological mechanisms underlying how squamate embryos respond to hypoxia remains poorly understood. Future research should characterize the physiological response of squamate embryos in utero and in ovo to hypoxia (e.g., lactate concentration, oxygen consumption, oxygen affinity of blood, and histology of extraembryonic tissues). Transcriptomic studies should characterize the gene expression response to hypoxia in utero and in ovo of viviparous and oviparous species to understand physiological plasticity of embryonic and maternal tissues in response to hypoxia. Potential target genes would include those in the hypoxia inducible factor 1 genetic network (Semenza, 2012; Wang & Semenza, 1993), which regulate the cellular response to hypoxia in animals. Beyond identifying gene expression mechanisms underlying the response to hypoxia, transcriptomic experiments could also identify likely genomic targets of selection.

5 | OTHER BIOPHYSICAL VARIABLES AND VIVIPARITY

This perspective should not be misread as dismissing the role of temperature in the evolution of viviparity, or as suggesting that oxygen availability is the only factor driving the evolution of viviparity. Like temperature, oxygen availability changes clinally with elevation in nature, which could create conditions for a selection gradient favoring “preadaptations” or “exaptations,” such as egg retention that could ultimately lead to viviparity (Blackburn, 1982). Indeed, the interaction of temperature and oxygen are likely important in shaping the selective landscape for the evolution of viviparity. For example, both ambient oxygen concentration and oxygen

affinity of hemoglobin decreases with increasing temperature (Dessauer, 1970; Hill et al., 2004; Pough, 1980). Hence, some of the lowest oxygen environments on earth are at high elevations in lower (and warmer) latitudes, which are precisely the areas with the highest relative frequencies of viviparous species within reproductively bimodal genera (Guillette et al., 1980; Hodges, 2004; Lambert & Wiens, 2013; Watson et al., 2014). Beyond oxygen, it is worth noting that the decrease in gas pressure at high elevations that drives elevational clines in oxygen availability can also impact other biophysical variables. Water vapor pressure also declines with increasing elevation, which would make eggs more susceptible to desiccation (Doody, 2009; Sexton & Marion, 1974). Cooler air at high elevations also tends to have lower humidity than warm air at low elevations, exacerbating water loss from eggs under low pressure at high elevation. This may be especially important for small squamate eggs. In fact, some lizards will retain eggs within their body in response to dry substrate (Andrews, 2018; Mathies & Andrews, 1999). Beyond better regulating temperature and oxygen availability, gravid females might also be able to ward off detrimental desiccation of embryos through viviparity (Doody, 2009).

6 | CONCLUSION

There is growing support in the biogeographic literature for the notion that oxygen availability might play an important role in the evolution of viviparity in squamate reptiles, but there is little physiological research on hypoxia and how it influences the developmental and reproductive physiology of oviparous and viviparous species. Future research programs should integrate studies of oxygen, temperature, and humidity on the development and physiology of oviparous and viviparous squamates to reveal the selective pressures that drive the evolution of viviparity.

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